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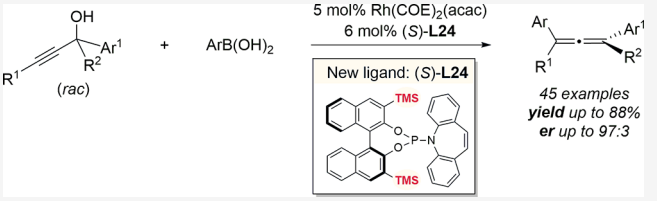
Catalytic Enantioselective S_N2'-Type Reaction for Axial Allene Chirality

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ABSTRACT: Optically active compounds are abundant in nature and are of fundamental interest for organic, medicinal, and material chemistry. Catalytic transformation of readily available racemic compounds into the same value-added enantiomeric products is one of the most efficient and cost-effective protocols. Herein, an S_N2'-type enantioselective reaction of racemic propargylic alcohols with aryl boronic acids (both are readily available) under a rhodium catalyst empowered by a new phosphoramidite−olefin bidentate ligand affording tetrasubstituted enantioenriched allenes has been unveiled. A pair of diastereomeric vinylic rhodium species with an allylic hydroxyl group has been identified and characterized by ESI-MS and TWIM-MS. Based on the X-ray diffraction data of the catalyst and other mechanistic studies, it is confirmed that the reaction is a conceptually new approach of parallel path enantio-convergent transformation, which is different from the classic dynamic kinetic resolution and dynamic kinetic asymmetric transformations.



INTRODUCTION

Optically active stereoisomers often elicit divergent biological responses owing to the intrinsic discriminatory capacity of biological systems, which in turn drives a perpetual demand for highly enantioselective synthetic methodologies toward the preferred stereoisomers.[1−3] During the last two decades, optically active compounds bearing central chirality or biaryl axial chirality have been extensively investigated due to their profound significance in biologically active natural and artificial products, chiral ligands, organocatalysts, and multifunctional synthetic building blocks.[4] As a sharp contrast, allenes exhibit a distinctly different type of axial chirality that extends across three contiguous carbons. Given the inherent steric distance between their 1,3-terminal substituents, the creation of this type of axial chirality is much more challenging and thus remains less developed.

Previously, optically active tetrasubstituted allenes[5] have been originally prepared by applying delicate precursors such as (di)enynones,[6] allenyl anion equivalents,[7] and 1,3-enynes.[8] Propargylic alcohols and derivatives are readily available and have been extensively applied as the versatile starting materials in highly efficient and selective syntheses.[9] In a seminal study, Jacobsen and co-workers developed the enantioselective allylation of tertiary propargyl acetates using chiral squaramide catalysts and TMSOTf as a promoter to construct all-carbon quaternary chiral centers via a catalytic enantioselective S_N1 process (Scheme 1a).[10] As a comparison, the free hydroxyl group in propargylic alcohols is a rather poor leaving group. The S_N2'-type substitution reactions of tertiary propargylic alcohols with a nucleophile would generate the optically active tetrasubstituted allene.

In 2017, we developed the transition metal-catalyzed *syn*-β−OH elimination reaction, which led to the (*S*)-tetrasubstituted chiral allenes from (*S*)-propargylic alcohols and *N*-methoxybenzamide.[11] Very recently, we reported a stereospecific Rh-catalyzed reaction of (*R*)-tertiary propargylic alcohols with organoboronic acids followed by the specific *anti*- or *syn*-β-hydroxyl elimination to afford (*S*)- or (*R*)-allenes by applying $[Rh(COD)OH]_2$ or $Cp^*Rh(OAc)_2$ as the catalyst, respectively (Scheme 1b).[12] The Pd-catalyzed kinetic resolution protocol for the asymmetric carboxylation of tertiary propargylic alcohols by using PPh_3 as the supporting ligand has also been developed for the construction of chiral tetrasubstituted 2,3-allenoic acids with the more slowly reacting isomer being recovered (Scheme 1c).[13] These transformations proceeded via the concept of either chirality transfer or kinetic resolution, whereas enantio-convergent synthesis from racemic tertiary propargylic alcohols remains a significant challenge and has not been established. Herein, we wish to report S_N2'-type reactions of racemic propargylic alcohols with another type of readily available and easy-to-handle stable chemicals, aryl boronic

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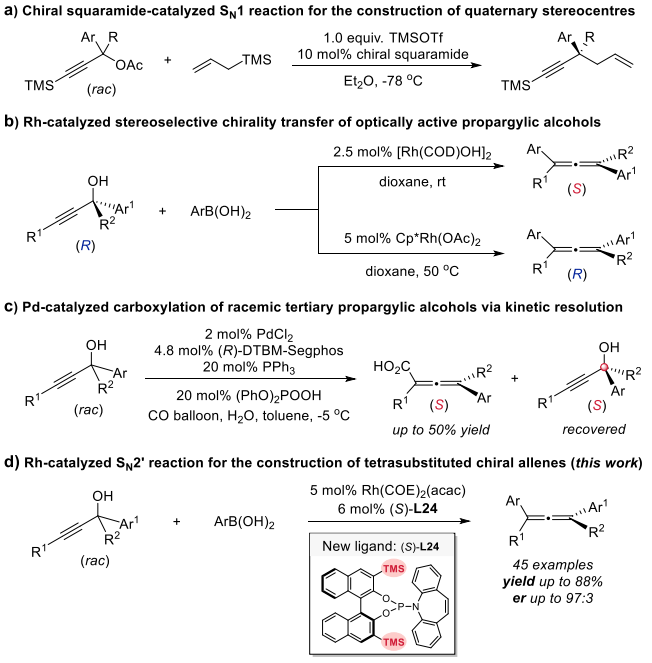


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Scheme 1. Selected Asymmetric Transformations of Propargylic Alcohols and Acetates



acids, for the enantioselective construction of tetrasubstituted chiral allenes enabled by the new phosphoramidite−olefin bidentate ligand (*S*)- or (*R*)-**L24** for rhodium catalysis (Scheme 1d). Mechanistic studies confirm this reaction as a parallel path enantio-convergent transformation (PPET),[14,15] differing from classic dynamic kinetic resolution and dynamic kinetic asymmetric transformations.

RESULTS AND DISCUSSION

Ligand Development

In order to have an enantio-convergent reaction taking the controlled *syn*- and *anti*-Rh−OH elimination observed in our previous study,[12] we started our study on the reaction of racemic 2-phenyloct-3-yn-2-ol *rac*-**1a** with 2 equiv of phenylboronic acid **2a** with 2.5 mol % $[Rh(COD)OH]_2$ and a chiral ligand as a catalyst by running the reaction in THF at 50 °C (Scheme 2a). Among the typical commercially available chiral biphosphine ligands (see Scheme S1 in Supporting Information for more detailed ligand screening), (*R*)-BINAP (**L1**),[16] (*S*)-SEGphos (**L2**),[17] (*R*)-Ph-Garphos (**L3**),[18] and (*R*,*S*)-Josiphos (**L4**)[19] afforded the product **3aa** in less than 10% yields with very poor ers. *P*,*N*-ligand (*R*,*R*)-^{i}Pr-Phosferrox (**L5**)[20] also did not work well. When (*S*,*S*,*S*)-Ph-SKP (**L6**),[21] (*R*)-Ph-SDP (**L7**),[22] or (*R*,*R*)-DIOP (**L8**)[23] was employed, the product **3aa** was obtained in an essentially racemic form; all these data indicate a similar level of energy for different pathways leading to both (*R*) and (*S*) enantiomers with similar efficiency. Both the (*R*,*R*)-DACH-Ph Trost ligand (**L9**)[24] and (*R*,*R*)-Ph-BPE (**L10**)[25] demonstrated no catalytic activity with 100% recovery of *rac*-**1a**. Further investigation on monophosphine ligands indicated the phosphoramidite ligand (*R*)-MonoPhos ((*R*)-**L11**)[26] could furnish the corresponding product **3aa** in 34% yield with 55:45 er. By switching the *N*,*N*-dimethyl moiety to 2-Me-THQ, i.e., (*R*)-**L12**,[27] the reaction afforded a 31% yield of racemic **3aa**. Employment of *N*-allyl phosphoramidite ligands (*R*)-**L13** and (*S*)-**L14**[28] provided better yields of 73% and 38%, respectively, but the er was still very low (54:46 er). To our delight, a 27% yield of **3aa** with a much improved er (71:29) was obtained when the phosphoramidite−olefin ligand (*S*)-**L15**[29] was used.

Considering the nature of low yield, we next increased the loading of the Rh catalyst to 10 mol % to further explore the chiral Rh catalyst effect as well as solvent and temperature with the promising ligand (*S*)-**L15** (Table 1). The yield of (*R*)-**3aa** was improved to 87% with 62:38 er when 10 mol % $[Rh(COD)OH]_2$ and 12 mol % (*S*)-**L15** were employed (Table 1, entry 1). Interestingly, the acetylacetonyl Rh catalysts, $Rh(acac)(COE)_2$, could furnish the desired allene (*R*)-**3aa** in 81% yield with 65:35 er (Table 1, entry 2; see Table S1 in Supporting Information for the detailed screening of the Rh catalyst), a promising result possibly following the concept of PPET. Reducing the catalyst loading to 5 mol % resulted in a decrease in the yield to 58%, while the er value of (*R*)-**3aa** was comparable (Table 1, entry 3). Then, we adjusted the molar ratio of phenylboronic acid **2a** over propargylic alcohol *rac*-**1a** to 1:2, and the er of (*R*)-**3aa** could be further improved to 78:22 (Table 1, entry 4), which indicated that the reactant stoichiometry was responsible for the improvement of the enantioselectivity. The temperature effect revealed the reaction at a lower temperature of 40 °C unexpectedly resulted in a diminished yield and er, while the reaction at a higher temperature (60 °C) effectively afforded the product (*R*)-**3aa** in 70% yield with 79:21 er (Table 1, entries 5 and 6). This is due to the higher reaction temperature, which accelerated the conversion of (*S*)-**1a** to (*R*)-**3aa** (for details, see Supporting Information). The solvent effect was subsequently evaluated: the reactions in dioxane gave (*R*)-**3aa** in 72% yield with 87:13 er (Table 1, entry 7; see Table S1 in the Supporting Information for the detailed screening of solvents). The reaction at a lower reaction concentration of 0.1 M contributed to better enantioselectivity, providing (*R*)-**3aa** in 64% yield with 91:9 er (Table 1, entries 8 and 9).

Encouraged by these data, we further optimized the backbone of the ligand for the purpose of improving the enantioselectivity (Scheme 2b). Besides binol-derived skeleton (*S*)-**L15**, semihydrogenated binol-derived skeleton (*S*)-**L16**, vanol-derived skeleton (*R*)-**L17**, and spinol-derived skeleton (*R*)-**L18** all failed to improve the enantioselectivity. Then, we explored the influence of different substituents at the 6,6′- or 3,3′-position of the binaphthyl skeleton: neither yield nor er was improved when 6,6′-dibromo- ((*S*)-**L19**) or 6,6′-diphenyl ((*S*)-**L20**) was applied. With 3,3′-dimethyl-substituted (*S*)-**L21**, the product (*R*)-**3aa** was obtained in 69% yield with an 87:13 er; no better results were observed when 3,3′-dibromo- or 3,3′- diphenyl-substituted ligands (*S*)-**L22** and (*S*)-**L23** were tested. To our delight, an 86% yield of (*R*)-**3aa** with 93:7 er was obtained upon applying the TMS substituent at the 3,3′-position of the binaphthyl skeleton ((*S*)-**L24**). Increasing propargylic alcohol *rac*-**1a** to 2.1 equiv afforded (*R*)-**3aa** in 79% yield with 95:5 er. Importantly, only trace amounts of the products were formed when the C=C bond hydrogenated analogue phosphoramidite ligand (*S*)-**L25** was employed, indicating a critical role of this C=C bond in the 7-membered ring of (*S*)-**L24**.

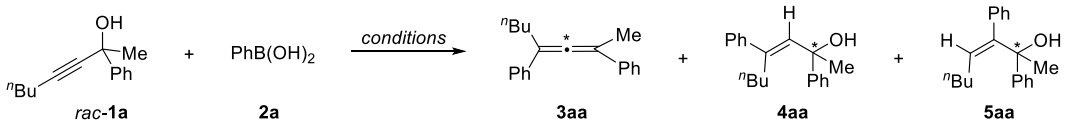
Identification of the Catalytically Active Species

In order to isolate the key catalytic active species for the reaction, we tried to synthesize complex Rh(*S*)-**L24**(acac) by mixing $Rh(COE)_2(acac)$ with ligand (*S*)-**L24** but failed.

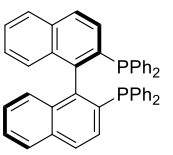
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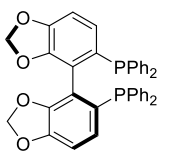
Scheme 2. Ligand Screening and Fine Tuning



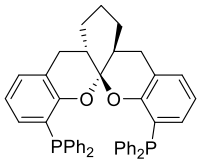
a) Ligand screening[a]



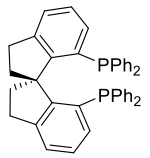
(*R*)-BINAP (**L1**)
NMR yield: 7% (54:46 er)
recovery of **1a**: 90%



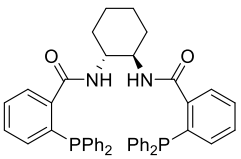
(*S*)-SEGphos (**L2**)
NMR yield: 3% (59:41 er)
recovery of **1a**: 96%



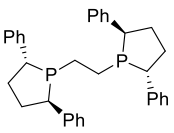
(*S,S,S*)-Ph-SKP (**L6**)
NMR yield: 63% (51:49 er)
recovery of **1a**: 30%
4aa: 1%, **5aa**: 5%



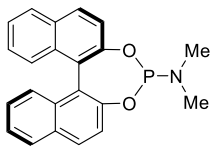
(*R*)-Ph-SDP (**L7**)
NMR yield: 54%(51:49 er)
recovery of **1a**: 34%
4aa: 1%, **5aa**: 5%



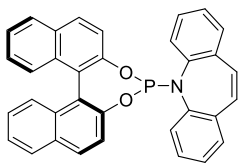
(*R,R*)-DACH-Ph
Trost Ligand (**L9**)
recovery of **1a**: 100%



(*R,R*)-Ph-BPE (**L10**)
recovery of **1a**: 100%

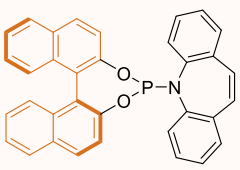


(*R*)-**L11**[b]
NMR yield: 34% (55:45 er)
recovery of **1a**: 55%
4aa: 9%

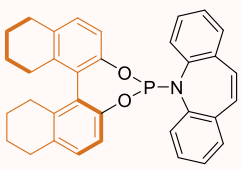


(*S*)-**L15**[b]
NMR yield: 27% (71:29 er)
recovery of **1a**: 63%
4aa: 4%

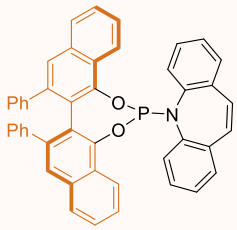
b) Ligand fine tuning[c]



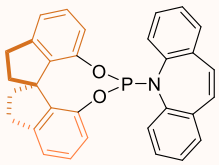
(*S*)-**L15**, 4 h
NMR yield: 64% (91:9 er)
4aa: 5%



(*S*)-**L16**, 7 h
NMR yield: 53% (72:28 er)
4aa: 3%, **5aa**: 4%

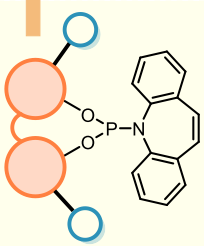


(*R*)-**L17**, 3 h
NMR yield: 80% (21:79 er)
4aa: 13%

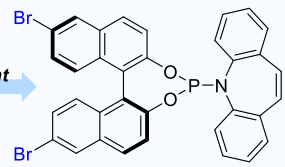


(*R*)-**L18**, 8 h
NMR yield: 58% (86:14 er)
4aa: 5%

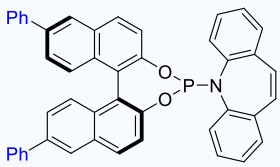
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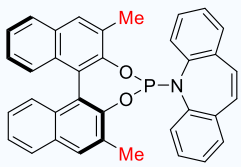
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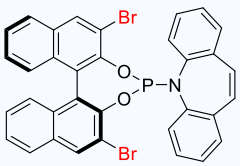
(*S*)-**L19**, 17 h
NMR yield: 46% (84:16 er)
4aa: 5%



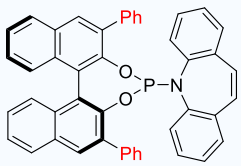
(*S*)-**L20**, 17 h
NMR yield: 28% (88:12 er)
4aa: 3%



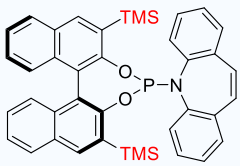
(*S*)-**L21**, 3 h
NMR yield: 69% (87:13 er)
4aa: 11%



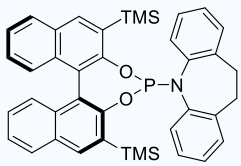
(*S*)-**L22**, 7 h
NMR yield: 42% (78:22 er)
4aa: 7%



(*S*)-**L23**, 11 h
NMR yield: 34% (80:20 er)
4aa: 7%



(*S*)-**L24**, 4 h
NMR yield: 86% (93:7 er), **4aa**: 9%
NMR yield: 79% (95:5 er), **4aa**: 8%[d]



(*S*)-**L25**, 10 h
4% yield

[a]Reaction conditions for ligand screening: *rac*-**1a** (0.2 mmol), **2a** (0.4 mmol), $[Rh(COD)OH]_2$ (2.5 mol %), and ligand (6 mol %) in THF (1.0 mL) at 50 °C for 5 h unless otherwise noted. All the yields were determined by ^1H NMR analysis using dibromomethane as the internal standard, and ee values were determined by HPLC analysis. [b]Ten mol % ligand was used. [c]Reaction conditions for ligand fine-tuning: *rac*-**1a** (0.4 mmol), **2a** (0.2 mmol), $Rh(acac)(COE)_2$ (5 mol %), and ligand (6 mol %) in dioxane (2.0 mL) at 60 °C for 5 h unless otherwise noted. All the yields were

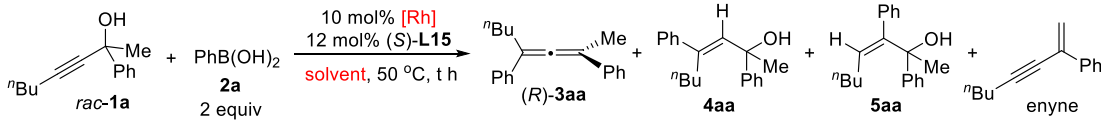
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Scheme 2. continued

determined by ^{1}H NMR analysis using dibromomethane as the internal standard, and ee values were determined by HPLC analysis. [d] *rac*-**1a**/**2a** (0.1 mmol) = 2.1:1, 5 h.

Table 1. Optimization of the Reaction Conditions[a]



entry	[Rh]	solvent	*t*/h	NMR yield[b] of (*R*)-**3aa**, er[c]/%	recovery[b] of *rac*-**1a**/%	NMR yield[b] of **4aa**/%	NMR yield[b] of **5aa**/%	NMR yield[b] of enyne/%
1	$[Rh(COD)OH]_2$	THF	6	87, 62:38	0	4	8	0
2	$Rh(acac)(COE)_2$	THF	2.5	81, 65:35	3	5	3	0
3[d]	$Rh(acac)(COE)_2$	THF	2.5	57, 66:34	34	6	3	0
4[d,e]	$Rh(acac)(COE)_2$	THF	4	60, 78:22	62	7	3	0
5[d,e,f]	$Rh(acac)(COE)_2$	THF	24	26, 70:30	82	4	0	0
6[d,e,g]	$Rh(acac)(COE)_2$	THF	3.5	70, 79:21	53	7	5	0
7[d,e,g]	$Rh(acac)(COE)_2$	dioxane	3.5	72, 87:13	49	8	6	0
8[d,g,h]	$Rh(acac)(COE)_2$	dioxane	4	64, 91:9	59	5	3	0
9[d,g,i]	$Rh(acac)(COE)_2$	dioxane	22	31, 87:13	70	3	0	0

[a]Reaction conditions: *rac*-**1a** (0.2 mmol), **2a** (0.4 mmol), [Rh] (10 mol %), and (*S*)-**L15** (12 mol %) in solvent (1.0 mL) at 50 °C unless otherwise noted. [b]Determined by ^{1}H NMR analysis using dibromomethane as the internal standard. [c]Determined by HPLC analysis. [d]$Rh(acac)(COE)_2$ (5 mol %) and ligand (6 mol %). [e]*rac*-**1a** (0.4 mmol) and **2a** (0.2 mmol). [f]40 °C. [g]60 °C. [h]Dioxane (2.0 mL). [i]Dioxane (3.0 mL).

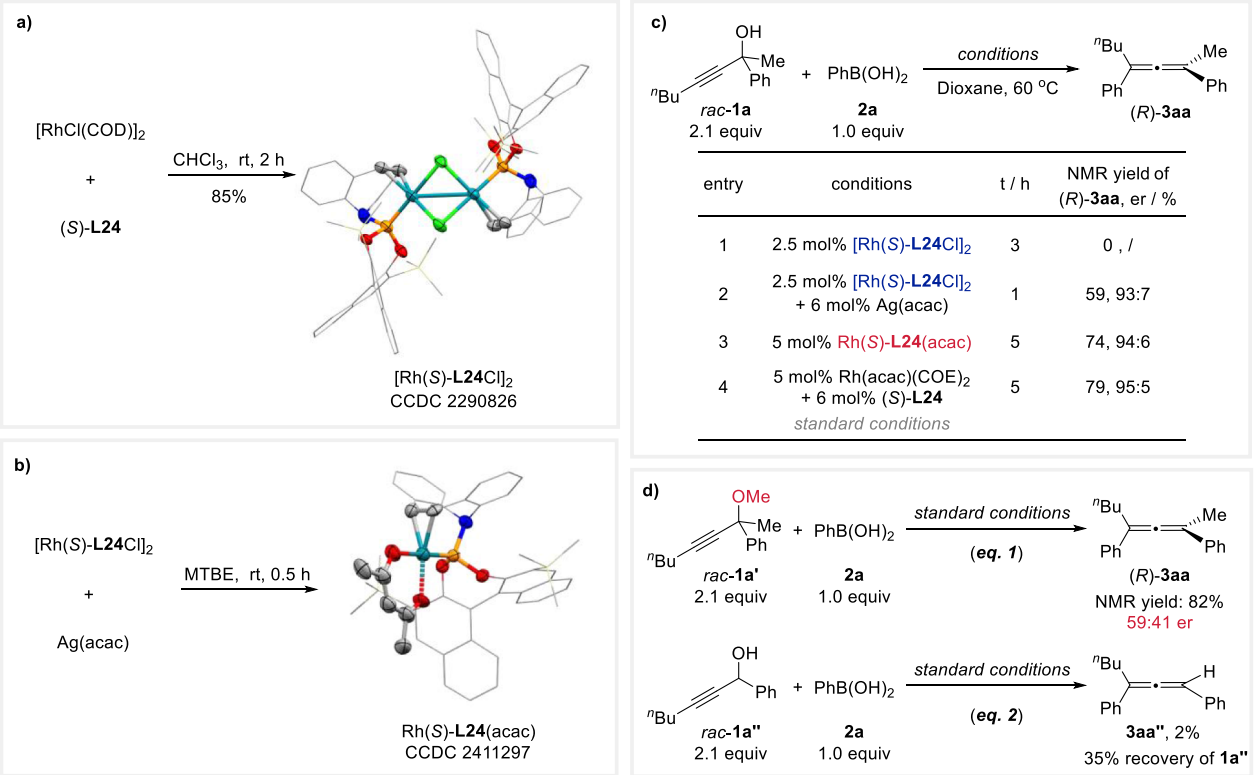


Figure 1. Mechanistic Studies. (a) Synthesis of $[Rh((S)\text{-}\mathbf{L24})Cl]_2$. (b) Synthesis of Rh(*S*)-**L24**(acac). (c) The reactions catalyzed by different Rh-species. (d) Control experiments.

Fortunately, after trial and error, single crystals of the Rh-dimer $[Rh(S)\text{-}\mathbf{L24}Cl]_2$ were obtained when $[Rh(COD)Cl]_2$ and (*S*)-**L24** were employed. Both the P atom and the C=C bond in the seven-membered ring of (*S*)-**L24** are coordinated to the Rh center and the two ligands located in the *trans* form (Figure 1a). However, there was no reaction with 100% recovery of *rac*-**1a** when this Rh-dimer was directly used as a catalyst under the standard conditions (Figure 1c, entry 1). To our delight, a 59% yield with 93:7 er of (*R*)-**3aa** was provided when 6 mol % Ag(acac) was added to break the Rh-dimeric nature of $[Rh(S)\text{-}\mathbf{L24}Cl]_2$ (Figure 1c, entry 2), which suggests that the dechlorinated monomeric Rh(*S*)-**L24**(acac) may be the

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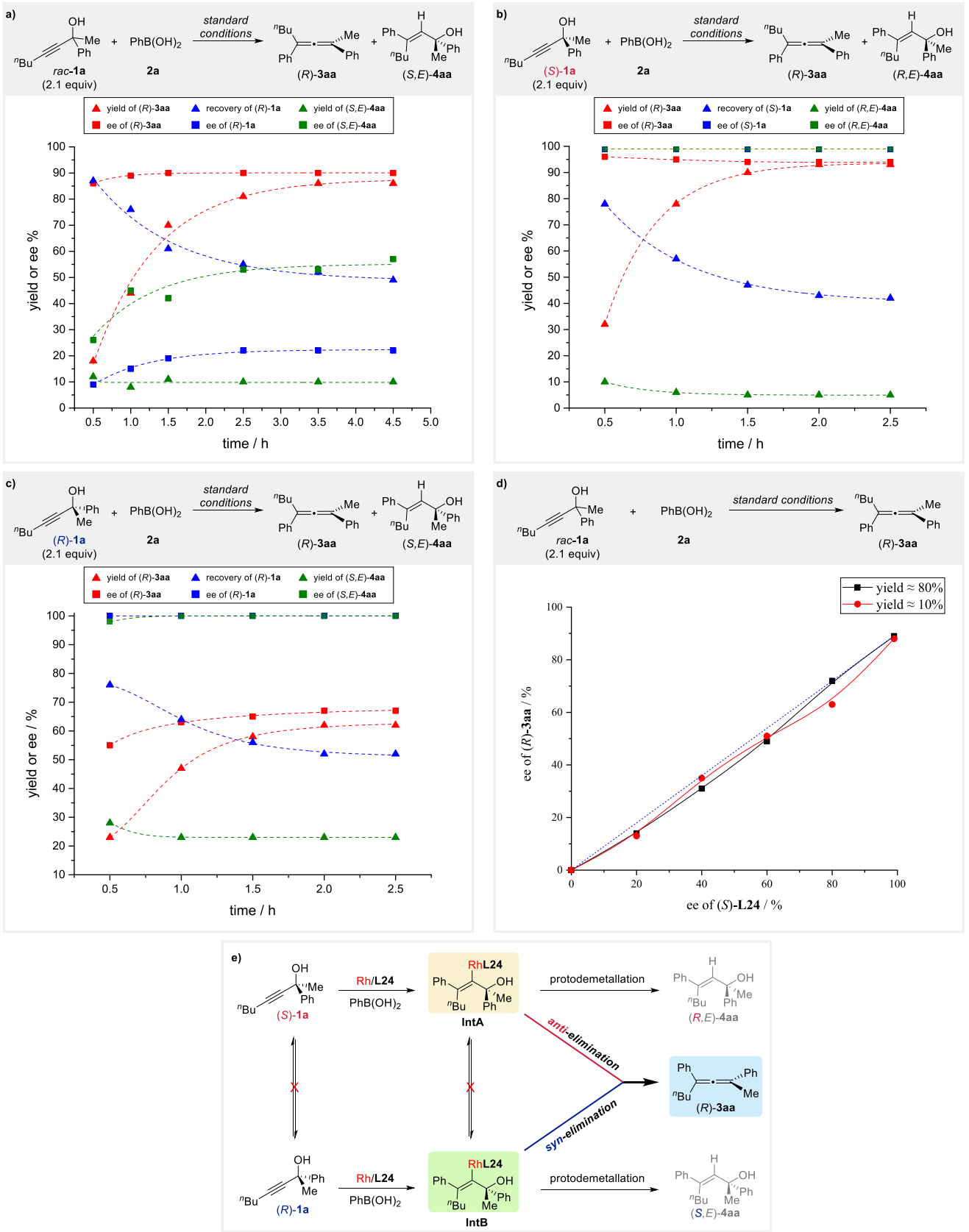


Figure 2. The reaction profiles, nonlinear effect studies, and a proposed mechanism. (a) The reaction profile of *rac*-**1a** under standard conditions. (b) The reaction profile of (*S*)-**1a** under standard conditions. (c) The reaction profile of (*R*)-**1a** under standard conditions. (d) The nonlinear effect studies. (e) The proposed reaction pathways. [#]The lines in Figure 1a–d are not trendlines and are included for visual aid only.

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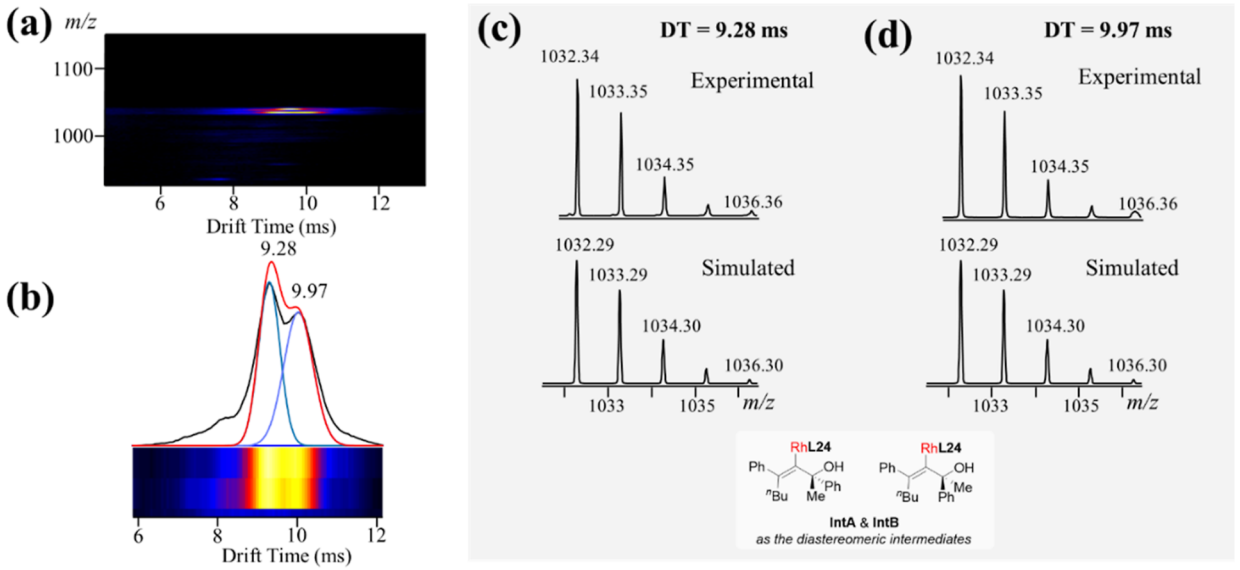


Figure 3. Multidimensional mass spectrometry. (a) Two-dimensional TWIM-MS plots collected on Waters Synapt G2-Si. (b) An expanded TWIM-MS plot of the ions alongside the chromatogram (black) and peak fitting curves (blue and purple) for $[\mathbf{IntA}\text{-H}]^{\bullet+}$ and $[\mathbf{IntB}\text{-H}]^{\bullet+}$. (c) Experimental isotope pattern of the ions with a DT of 9.28 ms and the simulated isotope pattern. (d) Experimental isotope pattern of the ions with a DT of 9.97 ms and the simulated isotope pattern.

catalytically active species for this reaction (for the detailed detections on the reaction of $[\text{Rh}(S)\text{-}\mathbf{L24}\text{Cl}]_2$ with Ag(acac), see Supporting Information). Finally, upon mixing $[\text{Rh}(S)\text{-}\mathbf{L24}\text{Cl}]_2$ with Ag(acac), Rh(*S*)-**L24**(acac) was successfully isolated and characterized by X-ray single-crystal diffraction study, further confirming that both the P atom and the C═C bond are coordinated to the Rh atom (Figure 1b). Indeed, the reaction with Rh((*S*)-**L24**)(acac) as a catalyst afforded (*R*)-**3aa** in 74% NMR yield with 94:6 er (Figure 1c, entry 3), which was similar to the data observed under the standard conditions (Figure 1c, entry 4), confirming that it is the catalytically active chiral catalyst for this reaction. Further control experiments showed that the reaction of propargyl methyl ether under the standard conditions afforded (*R*)-**3aa** in 82% yield with, however, an enantiomeric ratio of 59:41, indicating the importance of this hydroxy group in propargylic alcohol (Figure 1d, eq 1). The reaction of secondary propargylic alcohol *rac*-**1a″** under the standard conditions afforded trisubstituted allenes **3aa″** in 2% yield with 35% recovery (Figure 1d, eq 2), indicating the importance of the tertiary alcohol unit.

The Mechanistic Studies

In order to investigate the reaction mechanism, the reaction progress of racemic and chiral propargylic alcohol (*R*)-**1a** or (*S*)-**1a** had been monitored (Figure 2). Under the standard conditions, the reaction of *rac*-**1a** afforded the product (*R*)-**3aa** with high ee. The ee of recovered **1a** was only 22% with the (*R*)-isomer in excess at the time the reaction was complete (Figure 2a), which excluded the classic kinetic resolution process. The reaction was complete after 3.5 h; thus, the ee for **1a** levels off after that. When (*S*)-**1a** was applied (Figure 2b), the product (*R*)-**3aa** was obtained with >90% ee, and the ee of recovered (*S*)-**1a** remained unchanged (>99%) throughout the reaction. When (*R*)-**1a** was applied (Figure 2c), the reaction also yielded (*R*)-**3aa** as the major product, and the ee of recovered (*R*)-**1a** remained unchanged (>99%). All these data excluded the dynamic kinetic resolution process. Only the *E*-isomer of **4aa** was formed exclusively from either (*S*)-**1a** or (*R*)-**1a**, indicating no isomerization occurred to the in situ formed allylic C═C bond (Figure 2b,c), which excluded a DyKAT process via the possible *E* to *Z* isomerization of the C═C bond in the vinyl-rhodium intermediates formed from the insertion of the C−C triple bond of propargylic alcohol into the aryl−rhodium bond (for more information, see Supporting Information Figure S28). By comparison of Figure 2b with Figure 2c, a higher yield was observed in the reaction profile of (*S*)-**1a**. The conversion of (*S*)-**1a** was a lower barrier than that of (*R*)-**1a**, leaving a greater amount of (*R*)-**1a** unreacted, which resulted in an increase in its ee value (Figure 2a). A nearly linear effect was observed at both lower (∼10%) and higher conversions (∼80%), which suggested that the monomeric complex of Rh with (*S*)-**L24** may be the active catalytic species. It is consistent with our mechanistic studies that Rh(*S*)-**L24**(acac) is the active species for this transformation (Figure 1c).[30]

Based on these results, a unique mechanism has been proposed in Figure 2e. The *syn*-insertion of the C−C triple bond into the Ph-Rh**L24** species would lead to a pair of diastereoisomeric vinylic metallic species **Int A** and **Int B** (a pair of diastereoisomers), respectively, which would form the side product allylic alcohol (*R*)- or (*S*)-**4aa** upon protodemetalation.[31] Their subsequent *anti*- or *syn*-*β*−OH elimination for **Int A** or **Int B** in the same chemical environment, respectively, led to the same chiral allene (*R*)-**3aa**.

Now the challenge is to detect the key intermediates involved in the catalytic cycle. For such a purpose, multidimensional mass spectrometry of electrospray ionization mass spectrometry (ESI-MS)[32] and traveling-wave ion-mobility mass spectrometry (TWIM-MS)[33] has been employed to monitor the reaction of *rac*-**1a** (Figure 3). Ions with m/z = 1032 were detected (for more information, see Supporting Information). To obtain more structural information about these intermediates, TWIM-MS experiments, which are able to separate ions according to their various sizes and shapes, have been performed. As shown in Figure 3a,b, the captured intermediate ions with m/z = 1032 were resolved into two

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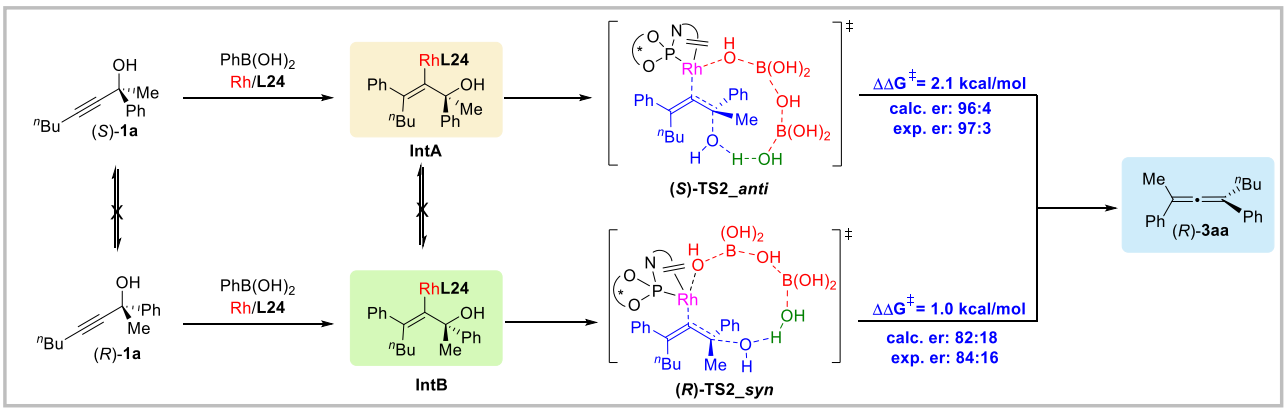


Figure 4. PPET mechanism with the enantioselectivity of each enantiomer validated by DFT studies.

peaks with drift times (DTs) = 9.28 and 9.97 ms. The experimental isotopic patterns of both signals matched the simulated isotope patterns for [**IntA**-H]$^{\bullet+}$ and [**IntB**-H]$^{\bullet+}$ (a pair of diastereoisomers) (Figure 3c,d).

Based on the results of DFT studies, we propose the mechanism outlined in Figure 4. Initially, regioselective *syn*-insertion of the C−C triple bond of propargylic alcohol (*S*)-**1a** or (*R*)-**1a** to the aryl-rhodium species formed from transmetalation of the Rh(I) catalyst with arylboronic acid gave the vinyl-rhodium **IntA** or **IntB**, which have been detected, separated, and characterized by MS studies (Figure 3). With the assistance of boric acid molecules and water, **IntA** may proceed with *anti*-*β*−OH elimination via the ten-membered cyclic transition state (*S*)-**TS2**_*anti*, while its diastereoisomer **IntB** generated from (*R*)-**1a** takes *syn*-*β*−OH elimination via the ten-membered cyclic transition state (*R*)-**TS2**_*syn*, both leading to the formation of the same enantiomer, tetrasubstituted allene (*R*)-**3aa**.

Substrate Scope

Having confirmed the PPET protocol and identified the optimal conditions, we next explored the generality of this Rh-catalyzed asymmetric coupling reaction between racemic tertiary propargylic alcohols and aryl boronic acids (Figure 5). Various commercially available aryl boronic acids were tested first. The model reaction with phenylboronic acid on a 0.5 mmol scale delivered the tetrasubstituted allene (*R*)-**3aa** in 81% yield with 95:5 er. The reaction of 3- or 4-methyl-substituted phenylboronic acids afforded (*R*)-**3ab** or (*R*)-**3ac** in 68% or 74% yield with 95:5 er, while the 2-methyl-substituted phenylboronic acid gave the product (*R*)-**3ad** in moderate er (78:22) most probably because of the steric effect. Other electron-donating groups, such as ethyl, *tert*-butyl, 3,5-dimethyl, phenyl, and 3-methoxy-substituted phenylboronic acids, all afforded the products (*R*)-**3ae**-**3ai** in decent yields of 45−71% with decent enantioselectivities (93:7−97:3 ers). 2-Naphthylboronic acid was also suitable to form the corresponding product (*R*)-**3aj**. The electron-deficient 4-bromo and 4- methoxycarbonyl-substituted phenylboronic acids also worked to give (*R*)-**3ak** and (*R*)-**3al** in 53% yield with 91:9 er and 50% yield with 92:8 er, respectively. Moreover, phenylboronic acid containing other synthetically useful or highly active functional groups including trimethylsilyl, vinyl, ethoxycarbonylethenyl, and bromomethyl also afforded the products (*R*)-**3am**-**3ap** in 44−79% yields with 93:7−96:4 ers.

Then, different tertiary propargylic alcohols with phenylboronic acid were evaluated. Electron-donating groups, such as methyl, methoxy, and isopropyl, and electron-withdrawing groups, including fluoro, chloro, bromo, 4-bromophenyl, and methoxycarbonyl, were well accommodated within the phenyl ring substituent of R^3 under the standard conditions to furnish the desired products (*R*)-**3ba**-**3ja** in good yields (65−88%) and decent enantioselectivities (91:9−95:5 ers). Besides ^{n}Bu (C_4), the scope of the R^1 group in tertiary propargylic alcohols is general since the substrates of R^1 with carbon chains of C_1, C_5, C_6, and even C_{14} as well as carbon chains bearing versatile functional groups, such as phenyl, allyl, chloro, cyano, internal C−C triple bond, free hydroxy, acetoxy, silyl ether, and carbazol-9-yl, could all afford the corresponding chiral tetrasubstituted allenes (*R*)-**3ka**-**3xa** in 33−88% yields with 90:10−95:5 ers. The alkyl R^2 may be further extended from methyl to ethyl and propyl to furnish (*R*)-**3ya** and (*R*)-**3za** in 75% yield with 95:5 er and 63% yield with 94:6 er. It is worth mentioning that propargylic alcohols bearing complex molecular units from (−)-perillyl alcohol, (−)-*β*-citronellol, and naproxen could also afford the corresponding products (*R*)-**3Aa**-**3Ca** in good yields and excellent diastereoselectivities (up to 96:4 dr), which showcased the generality and practicality of this strategy. The absolute configurations of (*R*)-**3ao** and (*R*)-**3ia** were determined to be *R* via single-crystal X-ray crystallography analysis, and other compounds were assigned accordingly. Notably, the gram-scale reaction also worked smoothly with (*R*)-**L24**, affording the opposite enantiomer (*S*)-**3wa** in similar yield and er, and the TBS could be easily removed with camphorsulfonic acid to afford allenol (*S*)-**3ua** in a nearly quantitative yield without erosion of enantiopurity; moreover, (*S*)-**3ua** could readily react with phthalimide or NBS to give the product *N*-allenyl phthalimide (*S*)-**6** or allenyl bromide (*S*)-**7** in excellent yields with a similar er; an oxidation reaction could furnish the allenyl aldehyde (*S*)-**8** in 80% yield with 94:6 er.

CONCLUSIONS

In summary, we herein observed a parallel path enantioconvergent transformation (PPET) for racemic alcohols: a rhodium-catalyzed asymmetric coupling reaction between readily available racemic propargylic alcohols and aryl boronic acids undergoes this unique path to access enantioenriched tetrasubstituted allenes incorporating various synthetically useful functional groups. Detailed experimental and DFT studies confirmed that the finetuned phosphoramidite−olefin

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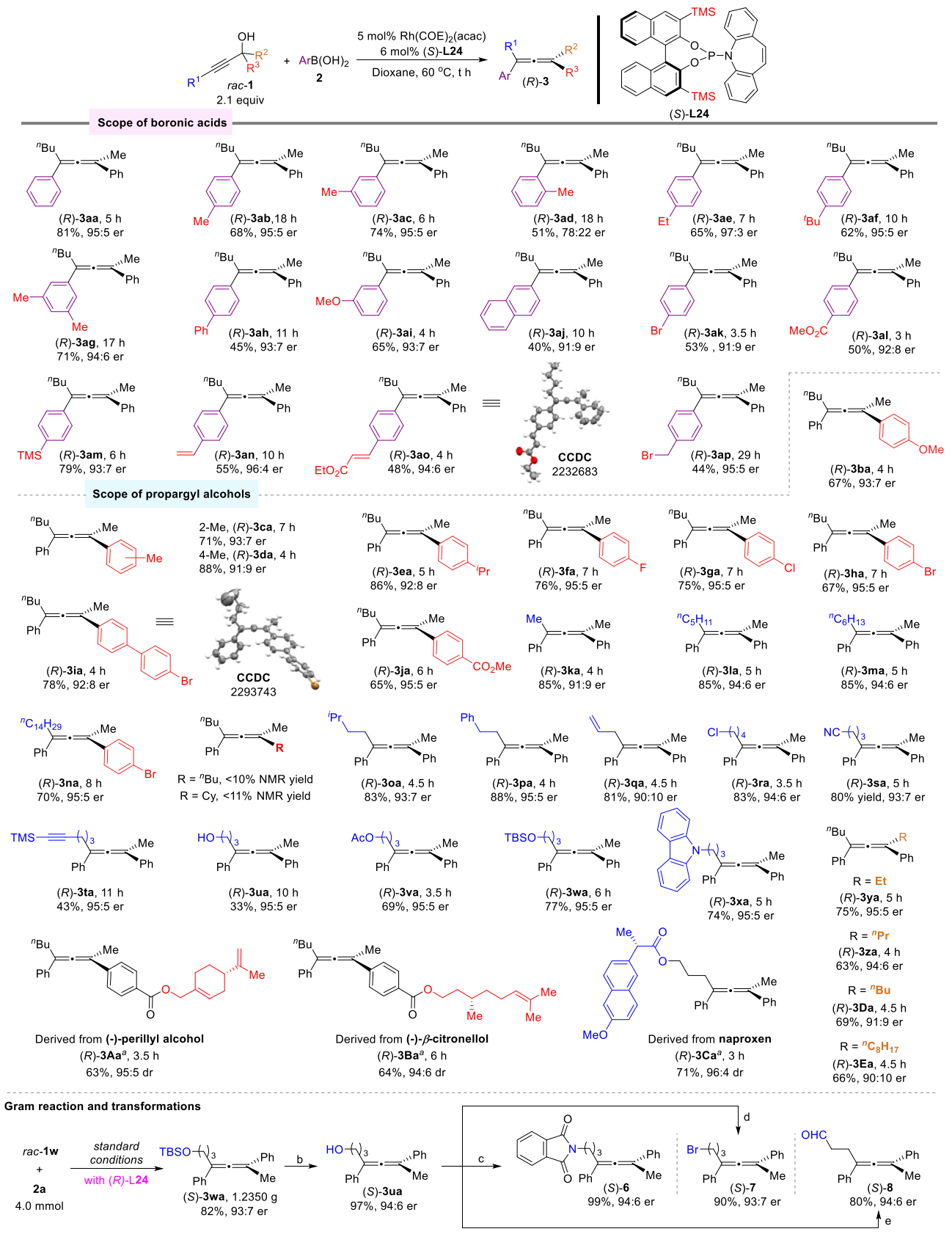


Figure 5. The substrate scope. Reaction conditions: *rac*-**1** (1.05 mmol), **2** (0.5 mmol), $Rh(COE)_2(acac)$ (5 mol %), and (*S*)-**L24** (6 mol %) in dioxane (5.0 mL) at 60 °C unless otherwise noted. (a) The reaction was conducted on a 0.15 mmol scale. (b) CSA (0.5 equiv), MeOH, rt, 1 h. (c) Phthalimide (1.2 equiv), PPh_3 (1.2 equiv), DEAD (1.2 equiv), THF, 0 °C to rt, 1.5 h. (d) NBS (1.5 equiv), PPh_3 (1.5 equiv), DMF, −10 °C, 6 h. (e) TEMPO (10 mol %), BAIB (1.1 equiv), DCM, rt, 5 h.

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ligand (*S*)-**L24** effectively coordinates to the rhodium center in a bidentate mode involving both the phosphine and olefin, which recognizes both enantiomers, and each resulting intermediate proceeds with parallel *anti*- or *syn*-Rh-*β*−OH elimination, respectively. Further studies of such a concept are currently underway in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.6c00454.

Detailed experimental procedures and characterization of new compounds Crystallographic information file for $[Rh(S)\text{-}\mathbf{L24}Cl]_2$ (CIF) Crystallographic information file for Rh(*S*)-**L24**(acac) (CIF) Crystallographic information file for (*R*)-**3ao** (CIF) Crystallographic information file for (*R*)-**3ia** (CIF) AUTHOR INFORMATION (PDF)

Accession Codes

Deposition Numbers 2232683, 2290826, 2293743, and 2411297 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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